

■ Scientific Justification (required for all)

Does an AGN’s infrared spectrum remember its accretion history? An active galactic nucleus (AGN) can undergo large luminosity changes within years [1,2], sometimes accompanied by a transformation of its continuum and broad-line spectrum [3–5]. These changing-look and strongly variable AGN let us ask how the surrounding material responds as nuclear activity rises, fades or recovers. Warm dust can remain bright after the observed nucleus fades, or lag behind renewed activity [6,7]. The unresolved issue is how strongly, and for how long, the infrared spectrum retains this earlier history, and whether delayed heating alone explains it.

We request **67.0 h of MIRI/MRS spectroscopy of 24 AGN at $z < 0.3$: 12 fading and 12 rising sources**. Every target has ZTF/WISE light curves, SPHEREx spectra and multi-epoch optical spectroscopy, including 2026 Palomar/NGPS spectra. Together, their recorded histories provide ensemble coverage of different stages of nuclear activity and dust response. MIRI supplies the essential warm continuum, both silicate features and spatial constraints on host emission, connecting each history to its warm-dust spectrum. The programme addresses two linked questions:

1. How strongly does the warm infrared spectrum retain recent accretion history? We will measure the warm-continuum excess or deficit relative to equilibrium at the current nuclear power, and its dependence on the direction, amplitude and age of the luminosity change.

2. Does this memory require a change in the dust distribution? Light-travel delays imprint history even on a fixed distribution [6,7]. We will test whether this response explains the spectra or whether the amount or location of the emitting dust must also change.

From infrared echoes to dust evolution. WISE observations have established infrared echoes of changing-look transitions [8–11], supporting intrinsic changes in nuclear power in those sources. Broadband colours, however, combine temperature, emitting area, extinction and host light [12–14]. The proposed spectra test the warm spectral signature of those histories and constrain which properties of the emitting material produce it.

The equilibrium sublimation radius scales approximately as $L^{1/2}$ [15–17], but the physical dust boundary can adjust over years [18]. Individual nuclei already suggest different responses: the smaller dust radius after Mrk 590 faded supports replenishment [19], whereas the temperature variations in Mrk 817 favour heating and cooling at a nearly fixed boundary [20]. These results motivate a comparison across both directions of change. Recovering sources further test whether the spectrum depends on the path to its present state.

MIRI’s diagnostic power. SPHEREx covers $0.75\text{--}5\text{ }\mu\text{m}$ [21], constraining hot dust close to the nucleus [20,22]. Longer delays and weaker variability at longer wavelengths [6,23,24] make the warm continuum sensitive to earlier illumination. MRS supplies the missing $4.9\text{--}27.9\text{ }\mu\text{m}$ spectrum [25]: the continuum measures the imprint, the paired silicate features constrain geometry and optical depth [26,27], and spatial profiles and aromatic bands constrain host emission [14,28,29]. At comparable current power, a warm excess in fading sources and a deficit in rising sources would connect spectral shape to the direction of change, with event-age trends testing its persistence. The result will measure, or bound, variability-induced biases in infrared estimates of nuclear power and dust covering factor [30,31].

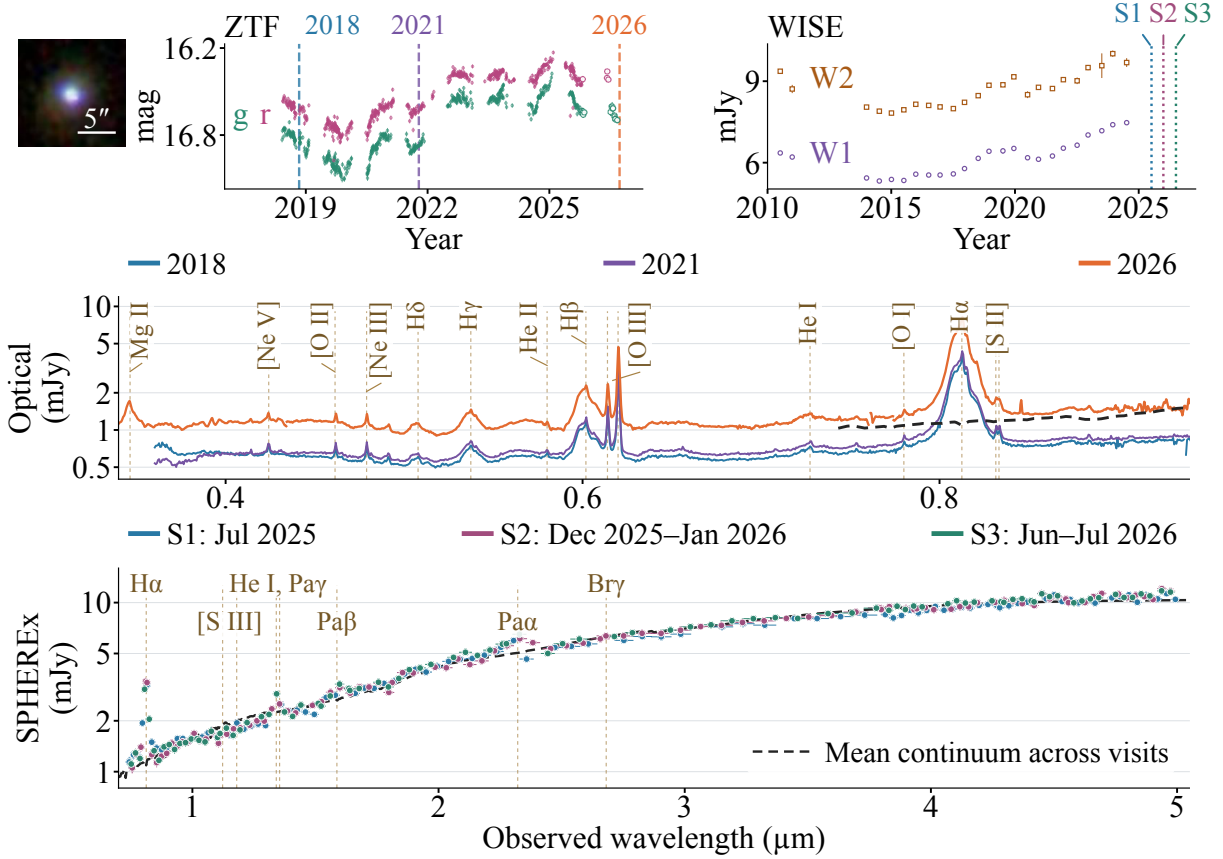


Figure 1: **Connecting nuclear variability to the hot-dust spectrum.** P9694 ($z = 0.238$): SDSS image ($5''$ bar), ZTF/WISE histories and optical/SPHEREx spectra. Labelled years on ZTF mark the optical spectra; open circles are ZTF alert photometry after the public release. WISE visit medians combine AllWISE and NEOWISE; S1–S3 mark the SPHEREx visits, and matching colours connect each epoch to its spectrum. The dashed host+disc+blackbody fit is the mean continuum across visits ($T_{\text{col}} \sim 1200$ K, model-dependent). Observed-frame wavelengths; fluxes in mJy on logarithmic axes without rescaling. Gold markers: expected line positions.

Tracking changes in AGN activity. We combine catalogued changing-look AGN [32–35] with large-amplitude ZTF/WISE variables [5,36], emphasizing variability direction, amplitude and timing across spectral classifications [33,37]. A variability manifold groups similar light-curve shapes to identify candidates [38]; multi-epoch spectra establish continuum and broad-line changes. The 12 rising and 12 fading selection histories span sustained changes, plateaus and recoveries. Each target has approximately a decade of W1/W2 monitoring; W1 95th-to-5th-percentile flux ratios span 1.3–2.3 before host subtraction. **This diversity tests infrared memory across different stages of change.** Recovering sources test whether nuclei at comparable current power retain different warm spectra because they followed different paths. SPHEREx and the 2026 NGPS spectra connect these histories to hot-dust emission and optical spectral state (Figure 1).

Measuring the spectral memory of accretion. Our primary observable is the logarithmic ratio of nuclear 8–13 μm continuum luminosity to its equilibrium prediction at current nuclear power (Figure 2). For each allowed dust model, this reference is the spectrum under steady illumination at the current luminosity. We will test history dependence jointly with luminosity, change amplitude and event age, allowing object-specific geometry, covering factor, extinction and host emission; the rising–fading offset provides a summary comparison. Comparisons within the overlapping luminosity range and with individual targets omitted will test its consistency.

Optical decomposition supplies the nuclear history [4], WISE constrains response amplitudes and measurable delays [24], and SPHEREx constrains hot dust at its observed epochs [20,22]. Rest-frame event ages are compared with measured lags or radius–luminosity priors [17,24,39], with warm delays inferred from each geometry [7]. The full recorded histories, including reversals, drive the predicted spectra [6,7]. We will marginalize over event dates, light-curve reconstructions across gaps and the conversion from optical to dust-heating luminosity, carrying these constraints into each target’s response stage.

Silicate profiles, spatial structure and aromatic bands constrain geometry and host emission [26,28]. Equilibrium and history-dependent models will share geometry and host priors; the history signal must survive these uncertainties. Mock MRS spectra spanning the allowed histories, nuclear fractions and geometries, with statistical and calibration covariance included, will pass through the same analysis to calibrate false-positive rates, detectable offsets and upper limits.

With twelve sources in each group, the mean-offset uncertainty is $0.408s$ for independent residual scatter s . For planning, $s = 0.20$ dex gives 0.082 dex precision and 80% power to detect a 0.24 dex offset (two-sided t -test, 5% significance). This conditional Gaussian estimate sets a sensitivity scale; the recovery tests extend it to the fitted covariates, correlated errors and individual sample uncertainties.

Testing for dust evolution. We will use clumpy and disc-plus-wind geometries [40–42] to construct responses to the reconstructed histories [6,7], allowing for composition and grain-size uncertainty [43]. Fixed distributions respond through grain heating and cooling [6]; evolving distributions also change the amount or location of dust [18,19,22]. Both include light-travel delays. The diagnostic

$$\delta_{\text{warm}} = \log_{10} \left(\frac{L_{8-13\mu\text{m}}^{\text{MRS}}}{L_{8-13\mu\text{m}}^{\text{fixed}}(t_{\text{MIRI}})} \right),$$

measures the warm-continuum residual relative to a *delayed fixed-distribution* prediction. The known limitations of static spectral models [28] motivate a joint test: evolving models must explain coherent continuum and silicate residuals across the allowed fixed models while also fitting the earlier data, with host and calibration uncertainties propagated throughout [44]. The hot-to-warm luminosity ratio provides a complementary test where the history constrains the 2–5 μm emission at the MIRI epoch.

Figure 3 compares delayed heating of fixed, optically thin shells with an illustrative limiting case in which the inner boundary adjusts immediately when changed illumination arrives. Where both hypotheses fit, we still measure or bound the warm spectral imprint of history.

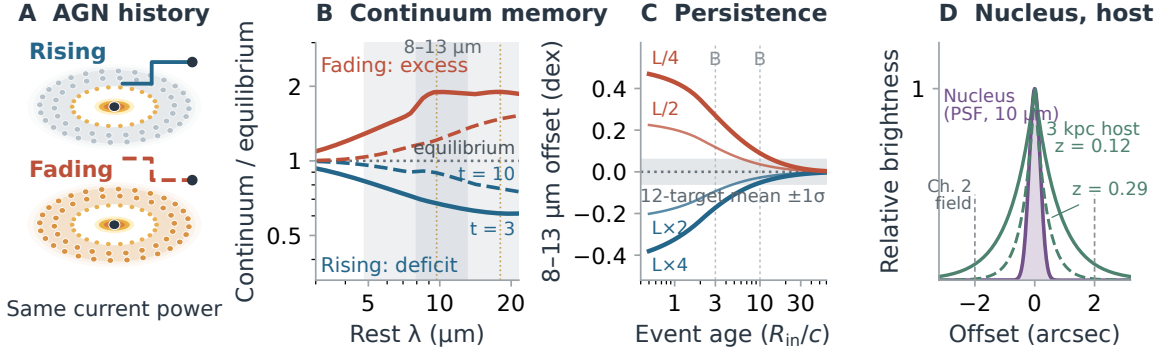


Figure 2: **Question 1: the spectral imprint of past activity.** A: equal current power, different histories. B: fixed optically thin dust with light-travel delays [6,45] after a fourfold rise (blue) or decline (red), divided by equilibrium at the current power, 3 and 10 inner-radius light-crossing times later; shading marks common MRS coverage and the 8–13 μm band. C: the 8–13 μm offset against event age for twofold and fourfold changes, with the $\pm 1\sigma$ precision of a 12-target mean (grey). D: MRS point-spread function at 10 μm [51] and a 3 kpc exponential host at the sample median and maximum redshift (dashed: channel 2 field [25]). B and C are illustrative, not target predictions.

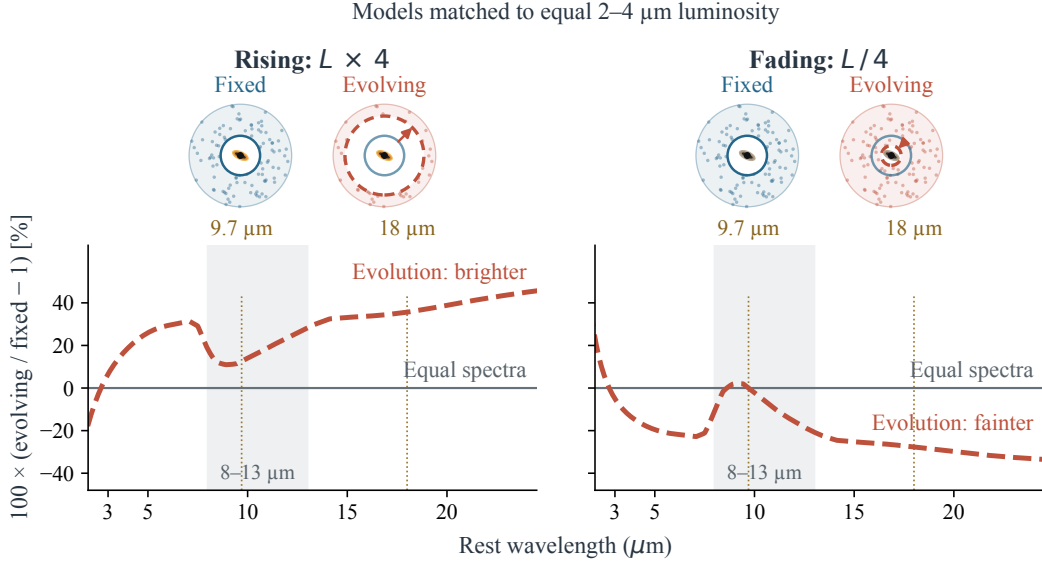


Figure 3: **Question 2: is dust evolution required?** Dashed curves show evolving-to-fixed spectral differences; zero denotes equal spectra. Shading marks the 8–13 μm diagnostic. Illustrative optically thin shells use Laor & Draine’s grain properties [45], a fourfold rise/decline and $R_{\text{in}} \propto L^{1/2}$ for the evolving boundary. Both models include light-travel delays. Spectra at $t = 20R_{\text{in,Si},0}/c$ (initial silicate inner radius $R_{\text{in,Si},0}$) are matched to equal 2–4 μm luminosity. Evolution changes the warm spectral shape even at equal hot-dust power.

■ Description of Observations (required for all GO and Survey)

Spectral coverage and sensitivity. We request one MIRI/MRS science visit and a matching sky visit per target. All three grating settings, nuclear target acquisition and a four-point dither provide approximately 4.9–27.9 μm coverage [25,46]. Both silicate peaks fall within this range for all 24 targets. The primary rest-frame 8–13 μm continuum diagnostic is accessible throughout the sample without channel 4C; the full spectrum constrains the longer-wavelength continuum and 18 μm profile.

Archival total-source W3 and W4 fluxes span 1.9–48.4 and 6.4–82.6 mJy. We target continuum S/N of 50 per $R = 100$ bin near rest 5 and 12 μm and 30 around the 18 μm feature, corresponding to 2% and 3.3% statistical errors per bin. We use FULL/FASTR1, one integration per dither, and 27/60/27 groups in settings A/B/C for 23 targets. Four dithers give 299.7/666.0/299.7 s on source, with all four channels observed simultaneously. The W3-faint J083826.50+371906.7 uses 80 groups in B (888.0 s). Equal-depth sky exposures preserve extended host emission for decomposition [46]. Science-plus-sky integration totals 17.00 h. **We request 67.0 h**, including acquisition, visit operations, slews and observatory overheads calculated with APT 2026.5.1: a Medium programme [47].

For J083826.50+371906.7 at $z = 0.2111$, ETC/Pandeia 2026.7 predicts S/N of 55/225/61 per $R = 100$ bin at rest 5/12/18 μm . These estimates assume a point source with $F_{\nu} \propto \lambda^2$, 0.95 mJy in W3 (half the archival total), the adopted sky spectrum, equal-depth sky exposures and extraction radii of 0.3"/0.6"/0.9". They establish statistical sensitivity under the adopted nuclear-flux model; host and calibration uncertainties enter the science inference separately. Acquisition uses F560W for the faintest sources and the neutral-density filter for brighter nuclei, with readouts giving S/N above 50 without saturation over the adopted flux ranges [48].

Connecting the observing epochs. The historical baseline gives each MIRI spectrum its temporal context. WISE and SPHEREx constrain the dust response at their observed epochs; optical spectroscopy tracks the nuclear spectrum. Optical monitoring near JWST anchors the nuclear continuum at the MIRI epoch. One MIRI visit per target connects recorded activity to the warm-dust response across 24 histories spanning different directions, amplitudes and event ages.

Separating nuclear dust from the host. Each MRS cube measures the nucleus and surrounding host together. Wavelength-dependent point-spread-function fitting separates unresolved and extended emission [28]; surrounding spaxels constrain the host spectrum directly. Joint modelling distinguishes continuum, silicate, PAH and H_2 contributions [13,49]. The 6.2, 7.7 and 11.3 μm PAH bands constrain star formation with allowance for AGN processing [29]; [Ne v] and [Ne vi] trace highly ionised gas [50]. Alternative extractions and host templates [14] define conservative nuclear-flux bounds for testing history dependence across the sample.

We will compare the 2010 W3/W4 photometry with redshifted nuclear-plus-host models integrated through the WISE bandpasses, including colour corrections and emission outside MRS coverage. Archival imaging constrains host light beyond the MRS field and within the SPHEREx aperture. Both science tests propagate nuclear-flux bounds and wavelength-dependent calibration covariance [44].

Supplemental Information

Special requirements. Each science/sky pair is linked as a consecutive, noninterruptible sequence to obtain a contemporaneous background [46]. Sky fields are selected using WISE and 2MASS catalogues and images, with at least $25''$ clearance from catalogued sources. Twenty-three fields lie within 1–2 arcmin of their nuclei; the crowded field around J20145928+2523010 uses a $135''$ offset. Background targets are marked extended and use four-point extended-source dithers, with no target acquisition. No fixed-date constraints, coordinated parallels or joint-facility time are requested. APT’s Visit Planner returns scheduling windows for all 48 visits.

Duplications. A coordinate-based MAST search on 2026 September 26, including archived and planned JWST observations, found no overlapping observations within $120''$ of any of the 24 science positions. No potential MIRI/MRS duplications were identified.

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